LLL-B-805

June 28, 1963

SUPERSEDING
Int. Fed. Spec. LLL-H-0060 (Army-CE)
May 14, 1959

FEDERAL SPECIFICATION

BUILDING BOARD, HARD PRESSED, VEGETABLE FIBER; PREFINISHED HARDBOARD PANELS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one grade of prefinished building board for interior use.

1.2 Classification. The prefinished building board shall be of the following classes, as specified (see 6.2):

Class A.—High gloss.
Class B.—Medium gloss.
Class C.—Low gloss.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following documents of the issues in effect on date of invitation for bids form a part of this specification to the extent specified herein:

Federal Specifications:

P-S-571—Soap, Scouring, Cake Form.
P-W-158—Wax, General Purpose, Solvent Type.
LLL-H-35—Hardboard, Fibrous-Felted (Fiberboard).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian agencies).
Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-W-3448—Wallboard; Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues in effect on date of invitation for bids shall apply.

Official Classification Committee Publication:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

American Trucking Associations, Inc., Publication:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N.W., Washington, D. C.)

3. REQUIREMENTS

3.1 Description. The prefinished building board shall be panels fabricated of hardboard as specified in 3.2 and coated as specified in 3.3. The size of the panels shall be as specified (see 6.2).

3.2 Hardboard. Hardboard shall conform to LLL-H-35, type II, treated or tempered.

3.3 Coating. The panels shall be thoroughly and evenly covered on one side with an undercoat and one or more coats of finishing material.

3.3.1 *Appearance of coating.* The coating shall be commercially free of orange peel, scratches, pinholes and pitholes, dirt specks, waviness, sagging, and uneven distribution. Fibers shall not be visible when the finish is illuminated and viewed at near-grazing angles.

3.3.2 *Color.* The color of the finish coating shall be as specified (see 6.2) and shall match the reference sample supplied by the purchaser or shall match a mutually agreed upon color chip.

3.3.3 *Film hardness.* The coating shall have a Sward Rocker value of not less than 30.

3.3.4 *Specular gloss.* The specular gloss (without correction for diffuse reflectance) shall be as follows:

Class A—not less than 85.
Class B—between 40 and 60.
Class C—not more than 39.

3.3.5 *Adhesion.* When tested as specified in 4.3.3.2, not more than 1 percent of 100 squares of the coating, cut 1/16 by 1/16 inch, shall be completely removed from the hardboard.

3.3.6 *Resistance to steam.* When tested as specified in 4.3.3.3, the coating shall show no raised fibers, swelling, blistering, loosening, or separation from the substrate, and the loss in gloss shall not exceed 4 units.

3.3.7 *Resistance to water immersion.* When exposed to cold distilled water for 96 hours as specified in 4.3.3.4, the coating shall show no raised fibers, swelling, blistering, wrinkling, or whitening immediately upon removal of the panel. After 18 hours of air drying, the gloss shall be within 3 units of that measured before immersion and the film hardness shall be no less than the value obtained prior to immersion.

3.3.8 *Resistance to loss of gloss, chalking, and color change.* When exposed and examined as specified in 4.3.3.5, the coating shall show no chalking, a loss of not more than 6 units of the original gloss, and a color change equivalent to a lightness index difference of not more than 2 units.

3.3.9 *Resistance to abrasion.* When tested as specified in 4.3.3.6, 35 liters of falling

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sand shall not abrade an area of the coating through to the substrate larger than 4 millimeters in diameter.

3.3.10 *Chemical resistance.* When tested as specified in 4.3.3.7, the coating shall withstand spot tests of the following solutions without showing any color change, softening, dulling, staining, blistering, or other defects:

2 percent by volume of reagent grade hydrochloric acid.
5 percent by volume of reagent grade glacial acetic acid.
3 percent by volume of reagent grade ammonium hydroxide.
1 percent by weight of reagent grade sodium hydroxide.
3 percent hydrogen peroxide.
95 percent denatured ethyl alcohol.
Xylene, A.C.S. reagent grade.

3.3.11 *Moisture content.* When tested as specified in 4.3.3.8, the loss in weight shall be not less than 3 percent.

3.3.12 *Resistance to temperature change.* When tested as specified in 4.3.3.9, the coating shall show no evidence of checking, cracking, crazing, blistering, or separation from the base material.

3.3.13 *Humidity resistance.*

3.3.13.1 *Tinted (colored) and patterned wallboard.* When tested as specified in 4.3.3.10, the coating shall show no evidence of raised fibers, blistering, swelling, peeling, or crazing, and the color shall match (by visual comparison) an unexposed panel.

3.3.13.2 *White wallboard (80 percent reflectance and above).* When tested as specified in 4.3.3.10, the coating shall show no evidence of film defects listed in 3.3.13.1 and the yellowness index difference shall be not more than 0.07.

3.3.14 *Scrub resistance.* When tested in accordance with 4.3.3.11, the difference in gloss before and after scrubbing shall not exceed the following:

Class A—Gloss loss of 10 units.
Class B—Gloss loss of 6 units.
Class C—Gloss increase of 5 units.

3.4 **Workmanship.** Workmanship shall be in accordance with commercial practice for this type of product. The prefinished hardboard shall be free of any defects that might impair its serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Inspection responsibility.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and test shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Classification of inspection.** Inspection shall be classified as follows:

(*a*) Acceptance inspection of the building board.
(*b*) Acceptance inspection of preparation for delivery of the panels.

4.3 **Acceptance inspection.**

4.3.1 *Sampling.*

4.3.1.1 *For examination.* Sampling for examination shall be in accordance with MIL-STD-105.

4.3.1.2 *For test.* Sampling for tests shall be in accordance with MIL-STD-105, inspection level L-3.

4.3.2 *Examination.* Samples selected in accordance with 4.3.1.1 shall be examined

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for the defects listed below. AQL shall be 1.5 percent defective.

101. Hardboard not as specified.
102. Color not as specified when examined in accordance with Fed. Test Method Std. No. 141, method 4250, paragraph 2.2.
103. Size not as specified.
104. Appearance of the panels not as specified.

4.3.3 *Tests.* Samples selected in accordance with 4.3.1.2 shall have specimens cut from each panel. Unless otherwise specified herein, the test specimens shall measure 24 inches by 48 inches for 4-foot-long samples and 24 inches by 54 inches for samples over 54 inches long. The test specimens shall be tested in accordance with the specified methods of Fed. Test Method Std. No. 141 and as specified herein. Any sample failing to pass one or more tests shall be considered a defective. Presence of one defective shall cause rejection of the lot represented by the sample.

4.3.3.1 *Film hardness.* Determine the film hardness with a Sward Rocker (see 6.3). Standardize the rocker by placing it on a clean polished plate glass and adjusting it to give exactly 50 oscillations. The number of complete oscillations multiplied by 2 gives the rocker value. Select a smooth, uniform area free from pits, bumps, or other surface defects. Operate the rocker on this area. Make three determinations on the coating, and determine the average. An average value of less than 30 shall constitute failure.

4.3.3.2 *Adhesion.* The following cross-cut method shall be used for determining adhesion of the coating. Cut 11 parallel lines into the coating in one direction, then cut 11 parallel lines at right angles to those previously cut. The cuts shall be spaced so as to have 100 squares measuring 1/16 by 1/16 inch. The cuts shall be made just deep enough to go through the coating but not into the board. Brush the cuts lightly with a fine brush. Examine the back of any square which is removed and disregard those with any hardboard attached. More than one square completely removed from the hardboard shall constitute failure.

4.3.3.3 *Resistance to steam.* Determine the gloss on a 6- by 6-inch test specimen using method 6101. Fill a 400-ml. beaker one-half full of water and place the test specimen, coated side down, on top of the beaker. Weigh down the test specimen with another 400-ml. beaker full of water. Heat the water in the bottom beaker to boiling and maintain a mild boil for 1 hour. Remove the panel and allow it to stand for 3 hours at room temperature. Examine the panel for defects and determine the loss in gloss. The following shall constitute failure: Raised fibers, swelling, blistering, loosening, separation from the substrate; or a loss in gloss of more than 4 units.

4.3.3.4 *Resistance to water.* Coat the back and edges of a 3- by 6-inch test specimen with an acrylic lacquer or similar material that will not soften or lose adhesion when immersed in water. Let dry. Determine the gloss of the face coating using method 6101 and determine the film hardness as specified herein. Immerse the specimen in distilled water and allow it to remain submerged for 96 hours at room temperature. Examine the specimen immediately upon removal from the water. The following shall constitute failure: Raised fibers; swelling, blistering, or whitening of the face coating. After 18 hours of air drying at room temperature, determine the loss of gloss and the film hardness. A change in gloss of more than 3 units or a film hardness less than that before immersion shall constitute failure.

4.3.3.5 *Resistance to loss of gloss, chalking, and color change.* Prepare another test

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specimen in accordance with 4.3.3.4. Determine the gloss using method 6101 and the daylight directional reflectance using method 6121. Subject the test specimen to 168 hours of accelerated weathering using method 6152. At the end of this period, examine the test specimen for chalking. Then wash the exposed test specimen under running water with a thoroughly degreased lamb's wool pad to remove all traces of dirt and scum. Wipe water off with clean cheesecloth and let dry for 2 hours. Determine the loss of gloss. Compare the color change before and after exposure by measuring the daylight directional reflectance in accordance with method 6121 and compute the lightness index difference in accordance with method 6122. The following shall constitute failure: Chalking; a loss of more than 6 units of the original gloss; or a color change equivalent to a lightness index difference of more than 2 units, an original gloss not as specified herein.

4.3.3.6 *Resistance to abrasion.* Using the apparatus and Ottawa sand described in method 6191, allow 35 liters of sand to fall on the coating of a test specimen. Remove the specimen and examine the abraded area. If the substrate is not visible or if the diameter of the area abraded through the coating to the substrate is not more than 4 millimeters, the coating shall have passed the abrasion test. If the diameter of the abraded area is more than 4 millimeters, perform the operation three more times. Average the diameters of the four abraded areas. An average diameter of abraded areas of more than 4 millimeters shall constitute failure.

4.3.3.7 *Chemical resistance.* Glass tubes for the following spot tests shall have the following nominal dimensions: Wall thickness—1/8 inch; inside diameter—7/8 inch; and depth—1/2 inch. The glass tubes shall be used to retain the chemical solutions on the coating. The following shall be performed: Seal the glass tubes to the coating of the hardboard with white Ceresine wax or similar material that will not be dissolved by the chemical solutions. Place 3 milliliters of the chemical solutions listed herein into the glass tubes spotted on the test specimen. Cover the tubes with glass and allow to remain at room temperature for 18 hours. At the end of this period, remove the solutions and the glass tubes. Remove the residual wax by rubbing gently with a cheesecloth and petroleum ether. Wash the test specimen with a mild soap solution, rinse in running water, let dry for 3 hours, and examine the coating. Any evidence of color change, softening, dulling, staining, or blistering of the coating shall constitute failure.

4.3.3.8 *Moisture content.* Weigh a 4- by 6-inch panel on an analytical balance. Place the panel in a well-ventilated oven and heat for 1-1/2 hours at a temperature of 290°F. (143°C.), cool in a desicator and reweigh. A loss in weight less than 3 percent shall constitute failure.

4.3.3.9 *Resistance to temperature change.* Place a 6- by 6-inch test specimen in a cold chamber and maintain the temperature at the specimen at minus 4°F. for 1 hour. At the end of this period, remove the test specimen and place it immediately in a ventilated oven. Maintain the oven temperature at 140°F. for 1 hour. Repeat for 5 cycles. At the end of the 5 cycles remove the test specimen and examine it. Any evidence of the coating checking, crazing, blistering, or separation from the base material shall constitute failure.

4.3.3.10 *Humidity resistance.* A 4- by 4-inch test specimen prepared in accordance with 4.3.3.4 shall be used in each of the following tests.

4.3.3.10.1 *Tinted (colored) and patterned hardboard.* Expose the test specimen in accordance with method 6132, except that it shall be exposed for 7 days. Any evidence of raised fibers, blistering, swelling, peeling, or crazing shall constitute failure. Mis-

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match in color (by visual comparison) between the exposed specimen and an unexposed sample shall constitute failure.

4.3.3.10.2 *White coated hardboard* (80 *percent reflectance and above*). Determine the yellowness index in accordance with method 6131 of the test specimen. Expose the test specimen as specified in 4.3.3.10.1. Redetermine the yellowness index after exposure and calculate the yellowness index difference in accordance with methods 6131. A yellowness index difference greater than 0.07 shall constitute failure.

4.3.3.11 *Scrub resistance.* The Gardner Straight Line Scrub Tester with the flat abrasion boat attachment is used for this test. Cut a test panel of the wallboard 6 by 17 inches. Determine the gloss of the coating (method 6101) and clamp the panel firmly in the pan of the apparatus. Cut a piece of new cheesecloth about 6- by 16-inch and fold to make a strip measuring approximately 6 by 2-1/4-inches. Soak the cheesecloth in distilled water and squeeze lightly. Then mount the strip of cloth in the abrasion boat. Pour 5 milliliters of distilled water over the cheesecloth pad. Rub a moist cake of grit soap, conforming to P-S-571, type I (Bon Ami), 10 cycles (i.e. 20 strokes—10 in each direction) across the pad using moderate pressure. Place the soap-coated pad on the coating, attach the abrasion boat to the machine and start the scrubbing action at approximately 40 cycles per minute. During the test, keep the panel just moist enough to allow the boat to move without drag by adding water dropwise in the path of the pad. After 100 cycles (200 separate strokes) remove the panel from the apparatus, rinse with running water, and allow to dry at room temperature for 2 hours. Then apply a coat of wax (conforming to P-W-158, type I) over the scrubbed surface and dry (about 1/2 hour). Buff the wax to its maximum luster and determine the gloss (method 6101) on the polished area. Express as units difference of gloss before and after scrubbing. A gloss loss of more than 10 units for class A specimens; a gloss loss of 6 units for class B specimens; or a gloss increase of more than 5 units for class C specimens shall constitute failure.

4.4 Acceptance inspection of preparation for delivery. The packaging and packing of the building board shall be examined to determine conformance to section 5 of this specification.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Packaging. The packaging of the building board shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* The building board shall be packaged in accordance with MIL-W-3448.

5.1.2 *Level C.* The building board shall be packaged in accordance with the supplier's standard practice.

5.2 Packing. The packing of the building board shall be level A or C, as specified (see 6.2).

5.2.1 *Level A.* The building board shall be packed in accordance with MIL-W-3448.

5.2.2 *Level C.* The building board shall be packed in containers complying with Uniform Freight Classification Rules or with National Motor Freight Classification Rules or with other carrier rules applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military agencies.* Packages and shipping containers shall be marked in accordance with MIL-STD-129.

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5.3.2 *Civil agencies.* Packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The prefinished hardboard panels are intended for interior use in the construction, maintenance, and repair of buildings.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Class of prefinished building board required (1.2).

(*c*) Size of panel required (3.1).

(*d*) Color of coating required (3.3.2).

(*e*) Administrative provisions for inspection records (4.1).

(*f*) Level of packaging and level of packing required (5.1 and 5.2).

6.3 A complete description of the Sward Rocker and its operation is given in "Physical and Chemical Examination of Paints, Varnishes, Lacquers and Colors" by Gardner Laboratory, Inc. P. O. Box 5728, Bethesda 14, Maryland.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:

Army—E

Navy—Y

Air Force—MOAMA

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